AE4L GATE

Dhruv Kakade

v26.0.0.1

Contents

List of Tables

List of Figures

Chapter 1

Syllabus For Gate AE 2027

1.1 Maths

1.1.1 Linear Algebra

Vector algebra, matrix algebra, systems of linear equations, rank of a matrix; Eigenvalues and eigenvectors.

1.1.2 Calculus

Functions of single variable, limits, continuity and differentiability, chain rule, maxima and minima, Integration; Functions of several variables, partial derivatives, gradient, divergence and curl, directional derivatives; Line, surface and volume integrals. Theorems of Stokes, Gauss and Green.

1.1.3 Differential Equations

First order linear ordinary differential equations; Higher order linear ODEs with constant coefficients; Classification of partial differential equations; *Solution to:* wave equation, Laplace equation, heat equation using separation of variables methods.

1.1.4 Numerical Methods

Numerical solutions for linear and nonlinear algebraic equations by bisection, Newton-Raphson method; basic numerical differentiation; numerical integration by trapezoidal and Simpson's rules; linear regression and least squares method; linear interpolation.

1.1.5 Special Topics

Fourier series; complex numbers, analytic functions and Cauchy-Riemann equations. Basics of probability and statistics: Bayes theorem, mean, median and mode; variance; binomial, normal and Poisson distribution.

1.2 Aerodynamics

1.2.1 Basic fluid mechanics

Fluid kinematics, streamline, streakline, pathline; conservation laws: mass, linear momentum and energy (integral and differential form); dimensional analysis and dynamic similarity; incompressibility conditions; Newtonian fluids.

1.2.2 Incompressible Flow

Elementary ideas of viscous flows

Hagen-Poiseulle flow, Couette flow, basic concepts in boundary layers, boundary layer thickness. Two-dimensional potential flow theory: sources, sinks, doublets, point vortex and their superposition, Bernoulli's equation.

Airfoils

Airfoil nomenclature; aerodynamic coefficients: lift, drag and moment; Kutta-Joukoswki theorem; thin airfoil theory, Kutta condition, starting vortex;

Wings

Finite wing theory: induced drag, Prandtl lifting line theory; critical and drag divergence Mach numbers.

1.2.3 Compressible flows

Basic concepts of compressibility, one-dimensional compressible flows, isentropic flows, normal and oblique shocks, Prandtl-Meyer flow; flow through nozzles and diffusers.

1.2.4 Special Topics

Fanno flow; Rayleigh flow; pressure measurements using U-tube manometers and Pitot probe.

1.3 Gas Turbines

Basics of thermodynamics.

1.3.1 Aerothermodynamics of aircraft engines

Thrust, efficiency, range. Brayton cycle.

1.3.2 Engine performance

Ramjet, turbojet, turbofan, turboprop and turboshaft engines; after-burners.

1.3.3 Aerothermodynamics of non-rotating propulsion components

Intakes, nozzles.

1.3.4 Gas turbine combustor

Types and configurations; combustion, stoichiometric fuel-to-air ratio.

1.3.5 Turbomachinery

Axial compressors

Angular momentum, work and compression, characteristic performance of a single axial compressor stage, efficiency of the compressor and degree of reaction, multi-staging;

Centrifugal compressor

Stage dynamics, inducer, impeller and diffuser; Axial turbines: stage performance.

1.3.6 Special Topics

Turbine blade cooling, compressor-turbine matching.

1.4 Rocket Propulsion

Rockets thrust equation and specific impulse, rocket performance; multi-staging; chemical rockets; performance of solid and liquid propellant rockets.

1.5 Flight Mechanics

1.6 Orbital Mechanics

1.7 Strength Of Materials

1.8 Aircraft Structures

1.9 Mechanical Vibration and Aero-elasticity

1.10 Aptitude

Chapter 2

Reference Books

2.1 Fluid Mechanics

1. Fluid Mechanics: Fundamentals and Applications by Yunus A. Cengel, John M. Cimbala
2. Fluid Mechanics by Frank M. White
3. Fluid Mechanics by RK Bansal (for basics and more solved problems)

2.2 Gas Dynamics

1. Fundamentals of Gas Dynamics by Robert D. Zucker, Oscar Biblarz
2. Fundamentals of Aerodynamics by John Anderson
3. Modern Compressible Flow: With Historical Perspective by John Anderson
4. Fundamentals Of Compressible Flow by SM Yahya
5. Aerodynamics for Engineering students by E. L. Houghton, N. B. Carruthers
6. Gas Dynamics by Ethirajan Rathakrishnan

2.3 Strength of Materials

1. Strength of Materials by R.K. Rajput
2. Mechanics of Materials by James M. Gere
3. Strength Of Materials Parts I and II by Timoshenko S.
4. Strength of Materials by S. Ramamrutham

2.4 Vibrations

1. Mechanical Vibrations by S.S Rao
2. Dynamics of Structures by Jagmohan L. Humar
3. Mechanical Vibrations by V.P Singh

2.5 Structures

1. Aircraft Structures for Engineering Students By T.H.G. Megson
2. Analysis and Design of Flight Vehicle Structures by E. F Bruhn
3. Mechanics of Aircraft Structures by C. T. Sun
4. Analysis of Aircraft Structures: An Introduction by Bruce k. Donaldson

2.6 Aircraft Propulsion

1. Gas Turbine Theory by Cohen, H., Rogers
2. Gas Turbines by V. Ganesan
3. Aircraft Propulsion and Gas Turbine Engines by Ahmed F. El-Sayed
4. Aircraft Propulsion by Saeed Farokhi

2.7 Rocket Propulsion

1.

- Rocket Propulsion Elements by George P. Sutton - Propellants and Explosives by Naminosuke Kubota - Rocket Propulsion by K Ramamurthi

- Heat transfer basics - Heat and Mass Transfer by R.K. Rajput - Heat and Mass Transfer by JP holman

- Engineering Mathematics - Advanced Engineering Mathematics by Erwin Kreyszig - Advanced Engineering Mathematics by H. K. Dass - Higher Engineering Mathematics by B. S. Grewal

- Flight Mechanics - Aircraft Performance and Design by John Anderson - Flight Stability and Automatic Control by R. C. Nelson - Airplane Performance, Stability and Control by Courtland D. Perkins, Robert E. Hage - Introduction to Flight by John Anderson - Aerodynamics by Clancy

- Space Mechanics - Orbital Mechanics for Engineering Students by Howard D. Curtis - Fundamentals of Astrodynamics by Donald D. Mueller, Jerry White, Roger R. Bate - Spaceflight Dynamics by William E. Wiesel - Orbital Mechanics by John E. Prussing, Bruce A. Conway - Analytical Mechanics of Space Systems by Hanspeter Schaub, John L. Junkins

- Previous Year Questions - GK Publisher

¿ MIT OCW / NPTEL - Assignments and Problem Sets ¿ The main books in focus are discussed below (soon). All the books mentioned need not be used.

Chapter 3

Unit Conversions and Constants

Chapter 4

Maths

Chapter 5

Aerodynamics

Chapter 6

Gas Turbines

Chapter 7

Rocket Propulsion

Chapter 8

Flight Mechanics

Chapter 9

Orbital Mechanics

Chapter 10

Strength of Materials

Chapter 11

Aircraft Structures

Chapter 12

Mechanical Vibration and Aero-elasticity